

Sec. A (Attempted any two questions) M.M. 6x2

1. Solve $(1 + e^{x/y}) + e^{x/y} \left(1 - \frac{x}{y}\right) \frac{dy}{dx} = 0$

2. Solve $y_2 - 6y_1 + 9y = x^2 + 2e^{2x}$

3. Solve $\frac{d^2y}{dx^2} + \frac{1}{x} \frac{dy}{dx} = \frac{12 \log x}{x^2}$

Sec. B (Attempted any one question) M.M. 4

1. Solve by changing the independent variable

$$\frac{d^2y}{dx^2} - \frac{1}{x} \frac{dy}{dx} + 4x^2y = x^4$$

2. Solve $\frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + y = xe^x \sin x$